

DESCRIPTION AND DISCUSSION ON DCASE 2026 CHALLENGE TASK 4: SPATIAL SEMANTIC SEGMENTATION OF SOUND SCENES

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ABSTRACT

This paper presents an overview of the Detection and Classification of Acoustic Scenes and Events (DCASE) 2026 Challenge Task 4, Spatial Semantic Segmentation of Sound Scenes (S5). The S5 task focuses on the joint detection and separation of sound events in complex spatial audio mixtures, supporting many applications, such as immersive communication. First introduced in DCASE 2025, the S5 task continues in DCASE 2026 with key changes to better reflect real-world conditions, including allowing mixtures to contain multiple sources of the same class and to contain no target sources. In this paper, we describe the task setting, including updates to the evaluation metrics and dataset, and report the results of 31 submissions from 10 teams. The official access point for data and code is https://github.com/nttcs-lab/dcase2026_task4_baseline.

Index Terms— Sound event detection and separation, Semantic segmentation of sound scenes, DCASE, First-order ambisonics

1. INTRODUCTION

Spatial semantic segmentation of sound scenes (S5) refers to the task of identifying and separating individual sound events from complex spatial audio signals. Given a multi-channel mixture as input, S5 produces a set of estimated single-channel source signals, each associated with a corresponding sound event class label. This capability supports the development of technologies across a wide range of applications, including immersive communication, extended reality (XR) systems, and acoustic scene monitoring [1,2].

DCASE 2025 Challenge Task 4 (DCASE25T4) was the first challenge to feature the S5 task [3], focusing specifically on indoor sound events captured using first-order Ambisonics microphone arrays. The challenge received submissions from eight teams, totaling 24 systems, most of which surpassed the baseline and demonstrated notable improvements in both sound event detection and separation. Several new approaches were explored, including iterative schemes that refine predictions by alternating between label estimation and source separation to enhance overall performance.

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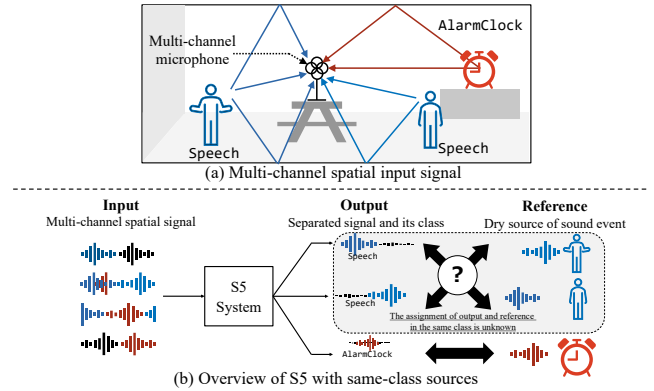


Figure 1: Overview of S5 task with same-class sources.

For the DCASE 2026 Challenge Task 4 (DCASE26T4), we organized a follow-up S5 task based on the DCASE 2025 setting, introducing two modifications to better reflect realistic and challenging conditions in real-world environments. First, in contrast to DCASE25T4, where labels within a mixture are mutually exclusive, DCASE26T4 allows duplicated labels. In other words, multiple sources from the same class can appear simultaneously within a single mixture. Such situations frequently occur in real acoustic environments, for example when multiple people talk simultaneously, or when mechanical fans blow from different directions, as illustrated in Fig. 1. This increases the difficulty of the task, as the ambiguity introduced by duplicated labels poses challenges for both the audio tagging and source separation components of most existing S5 systems. Second, in DCASE26T4, 16.7% of the evaluation mixtures contain zero target sound events, whereas in DCASE25T4 each mixture contains at least one target event. This setting is important for real-world deployments, where systems must operate continuously while target sound events occur only occasionally. The system must therefore not only detect when target sound events occur, but also correctly determine when they are absent. This becomes particularly challenging in the presence of background noise and interference sound events.

2. TASK SETTING OF S5

2.1. Formulation and notation

The S5 system receives as input $\mathbf{Y} = [\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(M)}]^\top \in \mathbb{R}^{M \times T}$, which denotes an M -channel time-domain mixture signal with du-

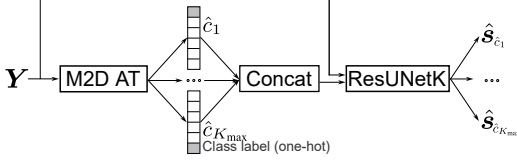


Figure 2: Overview of the baseline system.

ration T . Let s_{c_k} and s_{c_j} denote the target and interfering sound events, respectively; $h_k^{(m)}$ and $h_j^{(m)}$ denote their corresponding room impulse responses (RIRs); and $n^{(m)}$ denote the noise signal. The mixture at channel m can be synthesized as

$$y^{(m)} = \sum_{k=1}^K h_k^{(m)} * s_{c_k} + \sum_{j=1}^J h_j^{(m)} * s_{c_j} + n^{(m)}, \quad (1)$$

where K and J are the numbers of target and interference sound events in the mixture.

The output of the system consists of the estimated labels of the target sound events, $\hat{C} = (\hat{c}_1, \dots, \hat{c}_{\hat{K}})$, together with their associated separated single-channel waveforms measured at a reference microphone, $\hat{S} = (\hat{s}_{\hat{c}_1}, \dots, \hat{s}_{\hat{c}_{\hat{K}}})$, where $\hat{s}_{\hat{c}_i} \in \mathbb{R}^T$.

While the input and output are similar to those in DCASE25T4, the key difference is that the labels in C and $\hat{C}$ can be repeated. In other words, a mixture can include multiple sources belonging to the same class, each located at a different position in the room. The system is therefore expected to separate sources of the same class by leveraging differences in their spatial cues. In addition, K can range from 0 to $K_{\max}$, where $K = 0$ means no target sound event in the mixture.

2.2. Baseline system

Fig. 2 illustrates the two-stage baseline architecture [4], which combines Masked Modeling Duo (M2D)-based audio tagging (AT) [5] for sound event classification with ResUNet-based label-queried source separation (LQSS) [6]. Compared with the DCASE25T4 baseline [7], two key changes are introduced. First, the AT model now outputs $K_{\max}$ one-hot vectors rather than a single multi-hot vector to support same-class source detection. Second, while the ResUNetK architecture remains unchanged, the loss function is modified to handle cases where query labels are duplicated. Detailed training configurations, loss functions, and system specifications are available in [4] and DCASE26T4 baseline repository.

2.3. Evaluation metric

The class-aware permutation-invariant SDRi (CAPI-SDRi) [4] was adopted as the main ranking metric for DCASE26T4, which simultaneously evaluates the label prediction and source separation. The key idea is that estimated and reference sources are matched based on their labels. When the label is correctly predicted, SDRi is calculated on the waveforms, while for incorrect predictions, a penalty is applied. When multiple sources belong to the same class, a permutation-invariant objective is used to resolve the ambiguity.

For zero-target mixtures, there are no true positives (TP) or false negatives (FN). If false positives (FP) occur, the metric is computed normally. When there are no FP, i.e., a correct silence prediction, CAPI-SDRi is undefined. In such cases, the mixture is excluded

from the final averaging. Hence, correctly predicting silence contributes nothing, whereas false predictions are penalized.

It is worth noting that CAPI-SDRi is identical to CA-SDRi in DCASE25T4 when all source labels in the mixture are distinct, while additionally supporting mixtures with same-class sources. In addition to the main ranking metric, submissions are evaluated using other metrics, such as CASA-SDR [8], PESQ [9], and STOI [10], to provide complementary perspectives on performance.

3. DATASET FOR DCASE 2026 CHALLENGE TASK 4

We newly collected isolated target sound events, first-order Ambisonics RIRs, and background noise recordings. We combined these with screened recordings from the dataset released for DCASE25T4 and with selected publicly available data. These components were used to construct the dataset for DCASE26T4¹. Data were synthesized following the revised task setting, including mixtures with multiple same-class target sources and mixtures without any target sound events. The mixtures were generated with SpAudSyn², our spatial-audio synthesis tool developed for this task. The recorded components and the mixture generation procedure are described below.

3.1. Recorded components

As described in (1), each mixture is synthesized from target sound events, optional interference sound events, RIRs, and background noise. The recorded components used to construct these mixtures are summarized as follows.

- **Target sound events:** Table 1 summarizes the target sound-event samples used in the development and evaluation sets across 18 target classes. The development set combines 493 newly collected samples, 650 samples re-screened and selected from the DCASE25T4 dataset, and curated clips from publicly available datasets such as FSD50K [11], EARS [12], and Semantic Hearing [13] to enhance data diversity. The evaluation set consists exclusively of 626 newly collected samples. The newly collected recordings were acquired in an anechoic environment using three directional microphones arranged around a source area and one omnidirectional microphone placed above it. Human-produced classes were recorded from 4–8 participants, depending on the class, and object-based classes were recorded using 5–15 devices or object sets.
- **Room impulse responses:** In addition to the RIRs carried over from the dataset released for DCASE25T4 and from FOA-MEIR [14], new first-order Ambisonics RIRs were measured in six rooms, with two microphone-array placements per room, corresponding to a center position and a side position. All 12 placements are new environments not included in the DCASE25T4 dataset. The development and evaluation sets include the same six rooms but do not share any microphone-array placements. For each placement, a loudspeaker was placed every 20° over 18 azimuths on the horizontal plane, at elevations of 0° and ±20°, and at two source-array distances of 75 cm and either 120 or 150 cm depending on the room, yielding 1,296 RIRs in total. The RIRs were recorded with a Sennheiser AMBEO VR Mic and converted to B-format.

¹<https://doi.org/10.5281/zenodo.19328046>

²<https://github.com/nttclab/SpAudSyn>

Table 1: Statistics of target sound event recordings in the development and evaluation sets. The development set aggregates newly recorded data, DCASE25T4 recordings, and public datasets.

		Alarm Clock	Blender	Buzzer	Clapping	Cough	Cupboard OpenClose	Dishes	Doorbell	Foot Steps	Hair Dryer	Mechanical Fans	Musical Keyboard	Percu- sion	Pour	Speech	Typing	Vacuum Cleaner	Bicycle Bell	
Dev	new	duration [min]	3.5	1.7	1.2	3.7	0.2	2.5	2.0	3.0	12.0	10.8	30.7	4.2	9.5	1.1	2.7	0.4	14.8	1.1
		# of samples	26	7	17	45	2	37	20	37	91	8	21	39	57	11	39	7	5	24
	25T4	duration [min]	2.0	6.1	2.1	5.8	4.2	5.5	7.1	0.5	6.2	10.3	3.9	4.8	4.1	3.6	0.6	10.9	8.1	2.6
		# of samples	19	34	44	60	82	64	48	4	38	4	14	36	38	28	8	75	4	50
	public	duration [min]	8.5	11.1	3.2	10.7	20.6	17.1	9.1	8.1	29.5	6.1	17.3	44.4	78.1	5.5	66.8	43.3	10.4	5.6
		# of samples	42	52	21	312	211	184	208	63	298	25	59	397	1796	37	1195	228	46	54
Eval		duration [min]	4.6	6.8	4.5	4.1	5.4	2.1	2.5	2.5	6.1	18.5	17.8	4.5	7.5	5.5	2.4	7.2	17.0	1.5
		# of samples	33	60	48	19	88	25	23	32	32	8	7	34	30	62	30	68	4	23

- **Background noises:** Multi-channel environmental background noises were recorded with the same Sennheiser AM-BEO VR Mic and converted to B-format. The recordings may include naturally occurring directionality in the background noise field; for example, indoor noise levels may differ between the ceiling and horizontal directions. The recordings cover seven indoor and outdoor locations, with approximately 15 minutes recorded per location. Segments in which clear sound events or interfering sounds occurred near the microphone were removed during post-processing.

3.2. Mixture generation with SpAudSyn

The dataset released for DCASE25T4 was synthesized using a modified version of SpatialScaper [15]. SpatialScaper was originally developed to simulate and augment data for Sound Event Localization and Detection (SELD), and its original design does not explicitly provide dry reference signals required for source separation. To support the present task, we developed SpAudSyn for dataset construction. Its functional interface is inspired by SpatialScaper, but SpAudSyn provides flexible return options for labels, dry sources, metadata, and RIRs. Each sound scene can be fully parameterized and stored as a JSON file, which enables exact mixture reconstruction and reproducible experiments. The implementation also supports on-the-fly generation during training.

Each mixture in the dataset is synthesized by convolving target and interference sources with first-order Ambisonics RIRs and adding background noise as in (1). The mixture duration is 10 s, and all distributed mixtures are provided at 32 kHz/16-bit. The SNR of each target sound event is uniformly sampled between 5 and 20 dB relative to the background-noise level, whereas interference events are mixed at 0 to 15 dB. The maximum number of target sources in a mixture, $K_{\max}$, is limited to 3, all of which may belong to the same class. The maximum number of interference sound events is 2.

For mixtures with same-class target sources, the corresponding source directions are separated by at least 60° . This spatial information can serve as a clue for distinguishing sources. For mixtures with zero target sound events, only background noise and optionally interference sounds are present.

3.3. Development dataset

The development set is divided into three independent subsets: training, validation, and test. The training split provides isolated sources, RIRs, background noises, and interference sounds. Mixtures are generated on the fly during training. The validation split is distributed with fixed metadata so that the same mixtures can be reconstructed for model selection. The test split consists of fixed

synthesized mixtures for local evaluation before submission. The validation split contains 1,800 mixtures, and the test split contains 1,512 mixtures. In both splits, 16.7% of mixtures contain no target sound events, 16.7% contain one target sound event, 33.3% contain two target sound events, and 33.3% contain three target sound events. Within the two-target and three-target subsets, 50.0% of mixtures contain multiple same-class target sources.

3.4. Evaluation dataset

The evaluation dataset comprises 2,567 soundscapes, of which the first 1,512 synthesized soundscapes are used for ranking in DCASE26T4. These ranking soundscapes are generated using settings similar to those used for the test set in the development dataset, except that all components, including individual sound events, background noise, and FOA RIR, are newly recorded. The remaining soundscapes support task analysis through two subsets: synthesized mixtures whose components partly come from the training set, and real-world recordings. In the real-world subset, audio mixtures are recorded directly using an FOA microphone, the same microphone used to record the RIRs.

4. CHALLENGE RESULTS AND ANALYSIS

4.1. Overview of DCASE2026 T4 submissions

The challenge received 31 submissions from 10 teams [16–25]. Table 2 summarizes the architectural configurations, hardware specifications, and training strategies of the top-5 systems. Transformer-based backbones for source separation and extraction were widely used, often in combination with pretrained AT models such as M2D, BEATs [26], and ATST [27]. As in DCASE25T4, multi-stage (MS) training remained popular among most teams. In contrast, the first-ranked system adopted an end-to-end training strategy. The iterative refinement (IR) scheme, which alternates between label prediction and source separation, was also widely adopted, including by the top-4 systems [16–19]. Several systems made more effective use of spatial features or multi-channel inputs [19, 23, 25], while others focused on improving query reliability through tagger fusion [20, 22, 24], duplicate-label-aware source selection [21], or source counting [18]. For the loss functions, BCE and CE were the most common choices for label prediction and source counting depending on the model architecture, while the separation loss was mainly based on SNR (SDR) and its variants such as SA-SDR.

4.2. Performance of all submissions

Overall performance: Fig. 3 shows the performance of the best system from each team alongside the baseline system, evaluated on

Table 2: Technical characteristics of the top-5 systems in DCASE 2026 Task 4.

Rank	Model	# params.	GPUs	IR	MS	Acoustics feature	Data aug.	Loss
1	[16] TF-Locoformer, ATST-F	100M	32 × H200	✓		Spectrogram		BCE, SNR
2	[17] DeFT-Mamba, M2D, Audio-Flamingo 3	745M	4 × RTX 5090	✓	✓	Waveform, spectrogram	Difficulty-based mixup	SASDR, BCE, ArcFace, KL
3	[18] TF-Locoformer, BEATs, M2D, ATST-F, PE-A-Frame	687M	8 × A40	✓	✓	Spectrogram, Kaldi fbank, log-Mel spectrogram, DAC-VAE	Frequency warping, MixUp, filter aug.	SNR, CE
4	[19] TF-Locoformer, M2D, BEATs, ATST-F, fPaSST	375M	2 × RTX PRO 6000	✓	✓	Waveform, spectrogram, IPD/ILD spatial features	Frame shift, SpecAugment	BCE, SASDR, KL
5	[20] SR-CorrNet, M2D, PaSST	197M	2 × RTX 4090		✓	Spectrogram	SpecAugment, random gain filter, angle rotation	CE, SNR

Note: IR = Iterative refinement; MS = Multi-stage training.

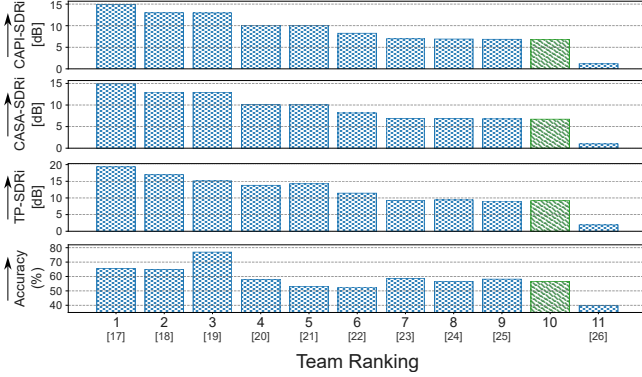


Figure 3: Performance of the best system from each team (blue dots) and the baseline system (green hatching), shown at their ranking positions.

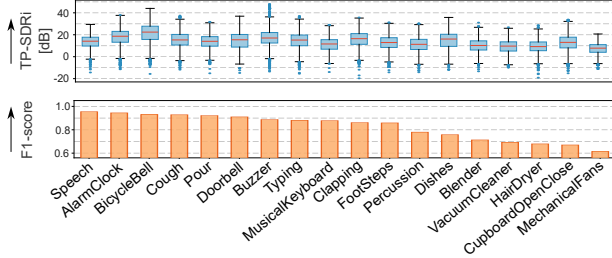


Figure 4: S5 performance per sound event class.

the main ranking subset of the evaluation dataset. Most systems outperform the baseline in the main ranking metric, CAPI-SDRi, and the top-ranked system achieves a notable margin. The CASA-SDRi [8], a variant metric that matches sources before accounting for classification errors, exhibits a similar trend. In contrast, TP-SDRi (measuring source separation) and Accuracy (measuring label prediction) reveal slightly different patterns. The third-ranked system attains the highest label prediction accuracy but lower source separation performance, resulting in an overall rank below the top two systems. In contrast, the fourth-ranked system is placed above the fifth-ranked system because its higher accuracy compensates for lower separation performance. These observations highlight the importance of jointly optimizing both label prediction and source separation to achieve strong performance on the S5 task.

Class-wise performance: Fig. 4 illustrates the TP-SDRi and F1-score of all systems across the 18 sound event classes. We can see that *Blender*, *HairDryer*, *MechanicalFans*, and *Vac-*

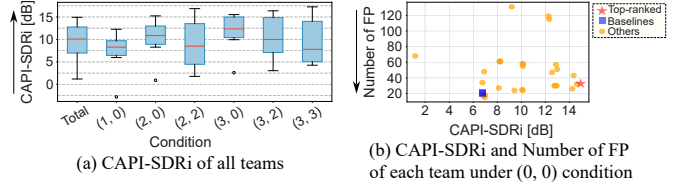


Figure 5: S5 performance under various label duplication conditions. The condition (K, D) indicates that the mixture contains K target sound events, of which D are duplicated labels.

uumCleaner show consistently low performance, as they produce spectrally similar continuous sounds that are difficult to distinguish from one another. *CupboardOpenClose* and *Dishes* also yield low F1-score, likely because their short and impulsive characteristics make them easily confused with other sound events. In contrast, classes with distinctive acoustic patterns, such as *AlarmClock*, *BicycleBell*, and *Speech*, achieve F1-score above 0.90 and high TP-SDRi.

Same-class performance: Fig. 5 (a) shows CAPI-SDRi of all systems under various conditions where multiple sources belong to the same class. Within the same number of target sound events, the performance decreases as the number of duplicated labels increases. This suggests that same-class sound events pose an additional challenge to the S5 task. Some top systems largely overcame this effect; for example, from $(3, 0)$ to $(3, 3)$, the baseline dropped by 5.40 dB, whereas the top system [16] instead improved by 1.57 dB.

Performance of zero target sound event case: Fig. 5 (b) shows the number of false positives (FP) in the zero-target event case $(0, 0)$. The top-ranked systems did not reduce these false detections compared with the baseline, and some produced even more. Reducing such false alarms in zero-target scenes therefore remains an unresolved issue and an important direction for future work.

5. CONCLUSION

This paper presented the task setting and results of DCASE26T4 on Spatial Semantic Segmentation of Sound Scenes. Compared with DCASE25T4, DCASE26T4 introduced more realistic conditions by allowing same-class target overlap and zero-target mixtures. The challenge received 31 submissions from 10 teams, with most systems outperforming the baselines. The improvements were largely associated with stronger model backbones, tighter coupling between detection and separation via iterative refinement, more effective use of spatial cues, and auxiliary mechanisms such as robust duplicate-label handling and false-positive suppression.

6. REFERENCES

- [1] M. Multus, S. Bruhn, J. Torres, E. Fotopoulou, T. Toftgård, E. Norvell, S. Döhla, Y. Gao, H.-y. Su, L. Laaksonen, *et al.*, “Immersive voice and audio services (IVAS) codec-the new 3GPP standard for immersive communication,” in *Proc. 157th Audio Eng. Soc. Conv.*, 2024.
- [2] A. Vasilache, T. Pihlajakuja, and M.-V. Laitinen, “Metadata-assisted spatial audio coding in IVAS codec,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*. IEEE, 2025, pp. 1–5.
- [3] M. Yasuda, B. T. Nguyen, N. Harada, R. Serizel, M. Mishra, M. Delcroix, S. Araki, D. Takeuchi, D. Niizumi, Y. Ohishi, *et al.*, “Description and discussion on DCASE 2025 challenge task 4: Spatial semantic segmentation of sound scenes,” in *Proc. Detect. Classif. Acoust. Scenes Events Workshop (DCASE)*, 2025.
- [4] B. T. Nguyen, M. Yasuda, D. Takeuchi, D. Niizumi, and N. Harada, “Class-aware permutation-invariant signal-to-distortion ratio for semantic segmentation of sound scene with same-class sources,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2026.
- [5] D. Niizumi, D. Takeuchi, Y. Ohishi, N. Harada, and K. Kashino, “Masked modeling duo: Towards a universal audio pre-training framework,” *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, 2024.
- [6] Q. Kong, K. Chen, H. Liu, X. Du, T. Berg-Kirkpatrick, S. Dubnov, and M. D. Plumbley, “Universal source separation with weakly labelled data for computational auditory scene analysis,” *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 34, pp. 206–219, 2025.
- [7] B. T. Nguyen, M. Yasuda, D. Takeuchi, D. Niizumi, Y. Ohishi, and N. Harada, “Baseline systems and evaluation metrics for spatial semantic segmentation of sound scenes,” in *Proc. Eur. Signal Process. Conf. (EUSIPCO)*, 2025.
- [8] M. Mishra, P. Magron, and R. Serizel, “Metric analysis for spatial semantic segmentation of sound scenes,” in *Proc. Eur. Signal Process. Conf. (EUSIPCO)*, 2026.
- [9] A. Rix, J. Beerends, M. Hollier, and A. Hekstra, “Perceptual evaluation of speech quality (PESQ)-a new method for speech quality assessment of telephone networks and codecs,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, vol. 2, 2001, pp. 749–752 vol.2.
- [10] C. H. Taal, R. C. Hendriks, R. Heusdens, and J. Jensen, “A short-time objective intelligibility measure for time-frequency weighted noisy speech,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2010, pp. 4214–4217.
- [11] E. Fonseca, X. Favory, J. Pons, F. Font, and X. Serra, “FSD50K: an open dataset of human-labeled sound events,” *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 30, pp. 829–852, 2021.
- [12] J. Richter, Y.-C. Wu, S. Krenn, S. Welker, B. Lay, S. Watanabe, A. Richard, and T. Gerkmann, “EARS: An anechoic full-band speech dataset benchmarked for speech enhancement and dereverberation,” in *Proc. Interspeech*, 2024, pp. 4873–4877.
- [13] B. Veluri, M. Itani, J. Chan, T. Yoshioka, and S. Gollakota, “Semantic hearing: Programming acoustic scenes with binaural hearables,” in *Proc. ACM Symp. User Interface Softw. Technol. (UIST)*, 2023, pp. 1–15.
- [14] M. Yasuda, Y. Ohishi, and S. Saito, “Echo-aware adaptation of sound event localization and detection in unknown environments,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2022, pp. 226–230.
- [15] I. R. Roman, C. Ick, S. Ding, A. S. Roman, B. McFee, and J. P. Bello, “Spatial scaper: a library to simulate and augment soundscapes for sound event localization and detection in realistic rooms,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2024, pp. 1221–1225.
- [16] Y. Bando, S. Sakurai, Y. Nozaki, K. Imoto, and M. Onishi, “End-to-end iterative S5 system based on TF-LoCoformer and ATST-Frame,” DCASE2026 Challenge, Tech. Rep., June 2026.
- [17] Y. Kwon, J. Park, H. Yin, and J.-W. Choi, “A multi-stage separation-and-classification framework guided by complementary acoustic-to-semantic clues,” DCASE2026 Challenge, Tech. Rep., June 2026.
- [18] K. Saijo, Y. Masuyama, C. Boeddeker, G. Wichern, J. Richter, T. Edo, and J. Le Roux, “The MERL systems for DCASE 2026 challenge task 4,” DCASE2026 Challenge, Tech. Rep., June 2026.
- [19] R. Wang, M. Chen, Y. Liu, L. Wu, and J. Chen, “Local-global transformer with iterative refinement for multi-channel sound source separation and extraction,” DCASE2026 Challenge, Tech. Rep., June 2026.
- [20] B.-H. Ku, W. Jeong, and H.-M. Park, “Extending SR-CorrNet to label-queried target sound extraction,” DCASE2026 Challenge, Tech. Rep., June 2026.
- [21] Y. You, “DCASE 2026 task 4 submission: Duplicate-label-aware source selection and TUSS overlay grafting,” DCASE2026 Challenge, Tech. Rep., June 2026.
- [22] S. Jeong and S.-E. Kim, “Tagging-driven inference refinement and dual-separator selection for spatial semantic segmentation of sound scenes,” DCASE2026 Challenge, Tech. Rep., June 2026.
- [23] S. Wang, C. Tang, Z. Zhang, and J. Yin, “Semantic distillation for spatial semantic segmentation of sound scenes,” DCASE2026 Challenge, Tech. Rep., June 2026.
- [24] Y. Deng, T. Zou, Y. Tian, H. Shi, Y. Yan, J. Luo, and G. Huang, “A label-guided ensemble system for spatial semantic segmentation of same-class sound sources,” DCASE2026 Challenge, Tech. Rep., June 2026.
- [25] Y. Park and H. Cho, “End-to-end spatial semantic separator with DOA module,” DCASE2026 Challenge, Tech. Rep., June 2026.
- [26] S. Chen, Y. Wu, C. Wang, S. Liu, D. Tompkins, Z. Chen, and F. Wei, “BEATs: Audio pre-training with acoustic tokenizers,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2023, pp. 5178–5193.
- [27] X. Li, N. Shao, and X. Li, “Self-supervised audio teacher-student transformer for both clip-level and frame-level tasks,” *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 32, pp. 1336–1351, 2024.